Supplementary Material for "Towards Universal AI-Generated Image Detection: A Systematic Literature Review of Methods, Generalization, Robustness, and Open Challenges"

Prasoon Kumar Kashyap, Aryan Jha and Soumyalatha Naveen

Journal: Artificial Intelligence Review
Authors: Prasoon Kumar Kashyap, Aryan Jha and Soumyalatha Naveen
Manipal Institute of Technology Bengaluru, Manipal Academy of Higher Education, Manipal, India
Corresponding authors: prasoonkkashyap@gmail.com; aryanjha.delhi@gmail.com; soumyalatha.naveen@manipal.edu

This document is Online Resource 1 of the article. It lists the study-level coding of all 99 included studies (Table S1), every transcribed performance value (Table S2), the reports excluded at full text (Table S3), the reports that could not be retrieved (Table S4) and the full Scopus search strings (Section S4). Study identifiers (R0001 to R0357) are the workbook Record_IDs.

S1 Study-Level Coding of the Included Studies

Task: Det = detection, Attr = attribution, Loc = localization, Bench = benchmark, Rob = robustness or attack study. Evidence (primary family, Axis B): SPA spatial, FRQ frequency, NOI noise/residual, SRC source/fingerprint, SEM semantic, REC reconstruction, GEO geometry/physical, HYB hybrid. Generators: G = GAN only, D = diffusion only, G+D = both, Multi = multi-family (more than the GAN and diffusion pair). Generalization tests (Axis D): U unseen generator, F cross-family, Ds cross-dataset, Dm cross-domain, O open set, C continual. Robustness (Axis E): R any test, S social-network processing, A adversarial. ✓ = tested; – = not tested; ? = unclear from the report. Str = strength of generalization evidence (VH very high, H high, M moderate, L low). QA = total appraisal score over QA1–QA8 (maximum 8); QA1, QA3 and QA8 were scored with AI assistance and validated by the authors (see the Quality Appraisal subsection of the main paper).

Table S1: Coding of the included studies

Study	Year	Short title	Task	Evid.	Gen.	U	F	Ds	Dm	O	C	R	S	A	Str	QA
[1] R0001	2026	A Multimodal Visionlanguage Model for Shrimp	Det	SEM	D	✓	–	–	–	–	–	–	–	–	H	4.5
[2] R0002	2026	PhyTrace	Det	GEO	Multi	✓	✓	✓	✓	–	–	✓	✓	–	H	7.0
[3] R0003	2024	Robust GAN-Based CNN Model as Generative AI Ap	Det	SPA	G	–	–	–	–	–	–	–	–	–	L	3.0
[4] R0004	2026	A Multi-Branch Transformer Based Cross-Attenti	Det	HYB	G+D	✓	✓	–	–	–	–	✓	–	–	M	6.5
[5] R0005	2026	Synthetic Image Detection Based on Multi-Level	Det	HYB	G+D	✓	✓	–	–	–	–	–	–	–	M	6.0
[6] R0006	2026	SCA-Det	Det	FRQ	Multi	✓	✓	✓	✓	–	–	✓	–	–	VH	6.5
[7] R0008	2026	MS-DCL	Det	HYB	G+D	✓	✓	✓	✓	–	–	✓	–	–	VH	6.5
[8] R0009	2026	FMACNN	Det	SPA	D	–	–	–	–	–	–	✓	–	–	L	5.5
[9] R0010	2024	IPD-Net	Det	SPA	Multi	✓	✓	✓	–	–	–	✓	–	–	VH	6.0

Study	Year	Short title	Task	Evid.	Gen.	U	F	Ds	Dm	O	C	R	S	A	Str	QA
[10] R0012	2026	Penny-Wise and Pound-Foolish in AI-Generated I	Det	SEM	Multi	✓	✓	✓	✓	–	–	–	–	–	H	6.0
[11] R0013	2026	Vulnerabilities in AI-Generated Image Detectio	Rob	FRQ	G+D	✓	✓	–	–	–	–	✓	–	✓	M	6.5
[12] R0014	2026	A robust synthetic face detector in OSN contex	Det	SPA	G+D	✓	–	–	–	–	–	✓	✓	–	M	6.5
[13] R0016	2025	Transfer Learning-Driven Hybrid Architecture f	Det	HYB	G	✓	–	✓	–	–	–	–	–	–	M	5.0
[14] R0017	2024	Generalized and robust model for GAN-generated	Det	NOI	G	✓	–	–	✓	–	–	✓	–	–	H	6.0
[15] R0018	2026	DCNet	Det	SPA	G+D	✓	✓	✓	✓	–	–	✓	–	–	H	6.5
[16] R0019	2022	An Eyes-Based Siamese Neural Network for the D	Det	SPA	G	✓	–	–	–	–	–	✓	–	–	H	6.5
[17] R0020	2026	SF-CFNet	Det	HYB	G+D	?	–	✓	✓	–	–	✓	–	–	M	6.0
[18] R0023	2026	Cross-Model Deepfake Detection Through Contour	Det+Attr	FRQ	G+D	✓	✓	✓	–	–	–	✓	–	–	M	6.0
[19] R0027	2025	Ensemble-Based Biometric Verification	Det	HYB	Multi	–	–	–	–	–	–	–	–	–	L	5.0
[20] R0029	2025	Detection of AI generated images using combine	Rob	HYB	G+D	✓	✓	✓	✓	✓	–	✓	–	✓	M	6.0
[21] R0031	2026	Revisiting DIRE	Det	REC	Multi	✓	✓	✓	✓	–	–	✓	–	–	VH	6.5
[22] R0032	2026	SDA-Det	Det	SPA	Multi	✓	✓	✓	✓	–	–	✓	–	–	VH	6.5
[23] R0034	2026	BM-DDFN	Attr	REC	Multi	✓	–	✓	✓	✓	–	✓	✓	–	M	7.0
[24] R0035	2024	A siamese-based verification system for open-s	Attr	SRC	Multi	✓	✓	–	✓	✓	–	–	–	–	H	6.5
[25] R0037	2024	Detecting images generated by diffusers	Det	SEM	D	✓	–	–	–	–	–	–	–	–	M	6.0
[26] R0039	2024	Boosting Deep Feature Fusion-Based Detection M	Det	SPA	G	–	–	–	–	–	–	–	–	–	L	4.5
[27] R0040	2026	Transformer Based on Multi-Domain Feature Fusi	Det	HYB	Multi	?	?	–	–	–	–	✓	–	–	L	3.0
[28] R0041	2026	VeriTrace	Bench	SPA	D	✓	–	✓	✓	–	–	–	–	–	M	5.5
[29] R0045	2026	Boon or Bane	Attr	SPA	D	–	–	–	–	–	–	✓	–	–	L	5.5
[30] R0046	2025	DiffCoR	Det	REC	Multi	✓	✓	–	✓	–	–	–	–	–	H	6.0
[31] R0047	2024	Artifact feature purification for cross-domain	Det	FRQ	G+D	✓	✓	–	✓	–	–	–	–	–	H	5.5
[32] R0051	2021	VIPPrint	Bench	SPA	G	–	–	–	–	–	–	✓	–	–	L	5.5
[33] R0052	2025	BOSC	Attr	SRC	Multi	✓	✓	✓	–	✓	–	✓	–	–	H	7.0
[34] R0053	2025	Advanced Detection of AI-Generated Images Thro	Det	SPA	G	–	–	–	–	–	–	–	–	–	L	4.5
[35] R0055	2026	SIDWA	Det	HYB	Multi	✓	✓	?	–	–	–	✓	–	–	H	6.5
[36] R0056	2026	Background-Noise-Driven Detection of Diffusion	Det	NOI	Multi	✓	–	✓	–	–	–	–	–	–	H	5.5
[37] R0058	2026	CLIP-ADA	Det	SEM	Multi	✓	✓	✓	✓	–	–	✓	–	–	VH	6.0
[38] R0062	2024	Enhancing Deepfake Detection With Diversified	Det	HYB	G	✓	–	✓	–	–	–	✓	–	–	M	7.0
[39] R0063	2026	A hybrid transfer learning approach for multi-	Det+Attr	HYB	G+D	–	–	–	–	–	–	–	–	–	L	5.5
[40] R0064	2025	Multi-modal texture fusion network for detecti	Det	HYB	Multi	✓	✓	✓	–	–	–	✓	–	✓	H	6.5
[41] R0065	2026	AI-generated image detection using Machine Lea	Bench	HYB	Multi	✓	✓	–	–	–	–	–	–	–	M	6.5
[42] R0066	2025	Vision Transformer-Based Framework for AI-Gene	Det	SPA	Multi	–	–	–	–	–	–	✓	–	–	L	4.5
[43] R0067	2026	An enhanced deep learning framework for DeepFa	Det	SPA	Multi	✓	?	✓	?	–	–	–	–	–	M	5.0
[44] R0069	2025	Sanitizing Diffusion-Generated Images via Fing	Rob	FRQ	D	–	–	–	–	–	–	✓	–	✓	L	6.0
[45] R0070	2024	Synthetic Face Discrimination via Learned Imag	Det	REC	G+D	✓	✓	✓	–	–	–	✓	–	–	H	7.0
[46] R0071	2024	Less is more	Det	FRQ	G	–	–	✓	–	–	–	✓	–	–	M	7.0
[47] R0072	2026	Geometric-semantic dual-adaptation for general	Det	SEM	Multi	✓	✓	–	✓	–	–	–	–	–	VH	6.0
[48] R0073	2025	Building a Universal Detector of AI-Generated	Det	HYB	Multi	✓	✓	✓	✓	–	–	✓	–	✓	H	7.0
[49] R0076	2025	Benchmark dataset and framework for continual	Det+Attr	SEM	Multi	–	–	–	–	–	✓	–	–	–	L	6.0
[50] R0077	2025	MFF-Net	Det	HYB	Multi	✓	✓	✓	✓	–	–	✓	–	–	VH	7.0
[51] R0078	2022	EXPOSING AI GENERATED DEEPFAKE IMAGES USING SI	Det	SPA	G	✓	–	✓	–	–	–	–	–	–	L	6.5

Study	Year	Short title	Task	Evid.	Gen.	U	F	Ds	Dm	O	C	R	S	A	Str	QA
[52] R0080	2026	Structure searchable network model for deepfak	Det	HYB	Multi	✓	✓	✓	–	–	–	✓	–	–	M	7.0
[53] R0081	2024	MMGANGuard	Det	SPA	G	–	–	–	–	–	–	–	–	–	L	4.5
[54] R0082	2026	Bias-Free? An Empirical Study on Ethnicity, Ge	Bench	SPA	Multi	✓	✓	✓	–	–	–	–	–	–	H	6.0
[55] R0083	2026	DTBF	Det	HYB	G+D	✓	✓	✓	–	–	–	–	–	–	H	5.0
[56] R0085	2024	Performance Comparison and Visualization of AI	Bench	SPA	Multi	✓	✓	✓	✓	–	–	✓	–	–	M	6.5
[57] R0086	2026	Linguistic profiling of deepfakes	Bench	SEM	Multi	–	–	–	–	–	–	✓	–	–	L	5.0
[58] R0089	2025	Fusing Frequency and Spatial Transformers for	Det	HYB	G	✓	–	–	–	–	–	–	–	–	M	6.0
[59] R0090	2026	LiteUpdate	Det	HYB	G+D	✓	✓	–	–	–	✓	✓	–	–	M	6.5
[60] R0091	2026	Generalizable and Adaptive Continual Learning	Det	SEM	Multi	✓	✓	✓	✓	–	✓	✓	–	–	H	5.5
[61] R0095	2025	DeepGuard	Other	SRC	D	–	–	–	–	✓	–	–	–	–	L	6.0
[62] R0096	2026	Harnessing attention for cropping and fusion i	Det	HYB	Multi	✓	✓	✓	–	✓	–	✓	–	–	H	5.5
[63] R0097	2026	NSFF	Det	HYB	Multi	✓	✓	✓	–	✓	–	✓	–	–	M	7.0
[64] R0098	2026	A Transfer-Learning Approach for Detection of	Other	HYB	Multi	–	–	–	–	–	–	–	–	–	L	5.5
[65] R0099	2025	Photo Authenticity Detection Using Machine Lea	Det	HYB	Multi	✓	–	✓	✓	✓	–	–	–	–	M	4.0
[66] R0101	2023	Black-box attack against GAN-generated image d	Other	SPA	G	✓	–	✓	–	✓	–	✓	–	✓	M	7.0
[67] R0104	2026	ADRD	Det	REC	Multi	✓	✓	✓	✓	✓	–	✓	–	–	M	7.0
[68] R0105	2024	Hierarchical Progressive Image Forgery Detecti	Det+Loc	HYB	Multi	✓	–	✓	?	–	–	–	–	–	L	5.0
[69] R0106	2025	SCADET	Det	FRQ	Multi	✓	✓	–	?	–	–	✓	–	–	H	7.0
[70] R0108	2023	AI vs. AI	Det	SPA	Multi	✓	–	✓	✓	–	–	–	–	–	M	6.5
[71] R0109	2026	Lightweight AI-Generated image detection based	Det	SPA	Multi	✓	✓	✓	✓	–	–	✓	–	–	H	7.0
[72] R0113	2026	Dual-path AI-generated image detection	Det	HYB	Multi	✓	✓	✓	✓	–	–	✓	✓	–	H	7.5
[73] R0114	2024	Reliable Out-of-Distribution Recognition of Sy	Det	FRQ	Multi	✓	✓	✓	✓	–	–	✓	✓	–	H	7.5
[74] R0115	2024	Unsupervised Generative Fake Image Detector	Det	SPA	G+D	✓	✓	–	–	–	–	✓	–	✓	M	7.0
[75] R0118	2026	Beyond Causal Models	Det	SPA	Multi	✓	✓	✓	–	–	–	–	–	–	H	5.5
[76] R0120	2025	Forensic Analysis of Manipulated Images and Vi	Bench	HYB	Multi	✓	✓	✓	–	–	–	–	–	–	M	6.0
[77] R0121	2026	Multi-Domain Perception Transformer for Genera	Det	HYB	Multi	✓	✓	✓	✓	–	–	–	–	–	H	6.5
[78] R0122	2025	Detecting AI-Generated Images Using a Hybrid R	Det	HYB	D	–	–	–	–	–	–	–	–	–	L	5.5
[79] R0126	2024	Synthbuster	Det	FRQ	D	✓	–	✓	✓	✓	–	✓	–	–	M	7.0
[80] R0128	2025	GCS-Net	Det	HYB	Multi	✓	✓	✓	✓	–	–	–	–	–	VH	6.5
[81] R0131	2025	Enhancing deepfake detection with Adaptive-DCG	Det	HYB	G	–	–	–	–	–	–	–	–	–	L	5.0
[82] R0133	2026	EXACT	Det+Attr	HYB	Multi	✓	✓	✓	✓	–	–	✓	–	–	H	7.0
[83] R0135	2026	Diffusion-Generated Image Detection via Target	Det	REC	G+D	✓	✓	✓	✓	–	–	–	–	–	M	6.5
[84] R0136	2026	Integrating semantic structure and texture cue	Det	HYB	D	–	–	–	–	–	–	✓	–	–	L	5.0
[85] R0144	2026	Generalizable Synthetic Image Detection via La	Det	SEM	Multi	✓	✓	✓	✓	–	–	✓	✓	✓	VH	7.0
[86] R0147	2026	Spatial flatness-curvature mask driven general	Det	SPA	Multi	✓	✓	✓	✓	–	–	✓	–	–	H	6.0
[87] R0148	2026	DIFC-Net	Det	HYB	Multi	✓	✓	✓	✓	–	–	–	–	–	H	6.5
[88] R0152	2025	Novel Deepfake Image Detection with PV-ISM	Det	SPA	Multi	–	–	✓	✓	–	–	–	–	–	M	5.5
[89] R0154	2025	RADAR	Det	HYB	D	–	–	–	✓	–	–	–	–	–	M	6.0
[90] R0155	2026	Color Space Analysis for AI-Generated Image De	Det	SPA	G+D	✓	✓	✓	–	–	–	✓	–	–	L	6.5
[91] R0228	2026	Proto-LeakNet	Attr	SPA	Multi	✓	✓	✓	✓	✓	–	✓	✓	–	H	7.5
[92] R0242	2026	The PRISM benchmark	Other	HYB	Multi	✓	✓	✓	✓	–	–	✓	–	–	H	6.5
[93] R0259	2025	How Much is the Source Mismatch an Important P	Other	SPA	Multi	✓	✓	✓	✓	✓	–	✓	–	–	M	6.5
[94] R0275	2022	FD2Foremer	Other	HYB	Multi	✓	✓	✓	–	–	–	✓	–	–	M	7.0
[95] R0302	2026	Toward Generalizable Forgery Detection and Rea	Det	HYB	Multi	✓	✓	✓	✓	–	–	✓	–	–	VH	6.5

Study	Year	Short title	Task	Evid.	Gen.	U	F	Ds	Dm	O	C	R	S	A	Str	QA
[96] R0309	2024	Detection of real-time deep fakes and face for	Other	SPA	G	✓	–	✓	–	–	–	–	–	–	M	6.0
[97] R0321	2025	DeepFeatureX-SN	Det+Attr	SPA	Multi	✓	–	✓	–	–	–	✓	–	–	M	7.0
[98] R0325	2025	DeepfakeCLIP	Other	SEM	G+D	✓	✓	✓	✓	✓	–	✓	–	–	H	6.0
[99] R0345	2023	Parameter agnostic stacked wavelet transformer	Other	FRQ	Multi	✓	✓	✓	✓	✓	–	✓	–	✓	H	7.0

S2 Reported Performance Values

Every performance value used in the Reported Performance Under Shift subsection of the main paper, as transcribed into workbook sheet 32 (197 values from 79 studies). Setting: in-domain, ood (out-of-distribution), open-set, robustness. Value in percent (values reported on a 0–1 scale are multiplied by 100). Stat: *mean* if the study reports an average over several targets, otherwise *value*. The workbook stores, for each row, the verbatim fragment of the extraction record from which the value was read. The authors validated every value.

Table S2: Transcribed performance values

Study	Setting	Shift	Target	Metric	Value	Stat
[1]	in-domain	none	same-pool split	AUC	100.00	value
[1]	ood	unseen-generator	withheld Midjourney v6	AUC	98.90	value
[2]	ood	multiple	ImageNet benchmark	AUC	81.98	value
[3]	in-domain	none	same-source split	ACC	96.35	value
[4]	in-domain	none	GenImage SD1.4	ACC	99.15	value
[4]	robustness	jpeg/blur mix	GenImage SD1.4	ACC	98.36	mean
[5]	in-domain	none	GenImage Midjourney	ACC	98.40	value
[5]	in-domain	none	GenImage Midjourney	AP	99.50	value
[5]	ood	unseen-generator	GenImage (7 unseen)	ACC	79.50	mean
[5]	ood	unseen-generator	GenImage (7 unseen)	AP	86.60	mean
[6]	in-domain	none	ProGAN	ACC	99.90	value
[6]	in-domain	none	ProGAN	AP	100.00	value
[6]	ood	multiple	45 generated-image sets	ACC	93.00	mean
[6]	ood	multiple	45 generated-image sets	AP	98.40	mean
[6]	robustness	jpeg	JPEG Q30	ACC	88.30	value
[7]	in-domain	none	GenImage SD1.4	ACC	99.90	value
[7]	in-domain	none	GenImage SD1.4	AP	100.00	value
[7]	ood	unseen-generator	GenImage	ACC	84.50	mean
[7]	ood	unseen-generator	GenImage	AP	97.30	mean
[8]	in-domain	none	RealAIGI/CIFAKE/FF++ splits	ACC	99.12	value
[9]	in-domain	none	ProGAN	ACC	99.94	value
[9]	ood	cross-dataset	GenImage	ACC	82.71	mean
[10]	ood	multiple	10 benchmarks	ACC	72.00	mean
[10]	ood	multiple	10 benchmarks	AP	86.14	mean
[12]	robustness	jpeg	JPEG Q50	ACC	98.29	value
[13]	in-domain	none	Kaggle source	ACC	97.93	value
[13]	in-domain	none	Kaggle source	AUC	98.80	value
[13]	ood	cross-dataset	FaceForensics++	ACC	94.18	value
[13]	ood	cross-dataset	Celeb-DF v2	ACC	91.87	value
[13]	ood	cross-dataset	DFDC	ACC	89.45	value
[13]	ood	cross-dataset	FaceForensics++	AUC	97.00	value
[13]	ood	cross-dataset	Celeb-DF v2	AUC	96.10	value
[13]	ood	cross-dataset	DFDC	AUC	94.90	value
[14]	in-domain	none	ProGAN (1-class)	ACC	98.00	value
[14]	in-domain	none	ProGAN (1-class)	AP	96.37	value
[14]	ood	unseen-generator	cross-GAN	ACC	80.83	mean
[14]	ood	unseen-generator	cross-GAN	AP	75.15	mean
[15]	ood	multiple	22 generators	ACC	90.78	mean
[16]	in-domain	none	StyleGAN2	AUC	100.00	value
[16]	ood	unseen-generator	ProGAN	AUC	99.70	value

Study	Setting	Shift	Target	Metric	Value	Stat
[16]	ood	unseen-generator	StyleGAN3	AUC	99.60	value
[17]	in-domain	none	SMTD	F1	98.96	value
[18]	ood	cross-dataset	leave-one-dataset-out	ACC	75.91	mean
[18]	ood	cross-dataset	leave-one-dataset-out	AUC	90.85	mean
[19]	in-domain	none	pooled six sources	ACC	79.20	value
[20]	in-domain	none	GenImage SD1.5	ACC	99.80	value
[20]	ood	unseen-generator	Midjourney	ACC	66.02	value
[20]	ood	unseen-generator	GLIDE	ACC	73.61	value
[20]	ood	unseen-generator	VQDM	ACC	57.17	value
[20]	ood	unseen-generator	BigGAN	ACC	55.97	value
[20]	ood	unseen-generator	StyleGAN3	ACC	45.34	value
[21]	ood	multiple	30 generators	ACC	96.50	mean
[21]	ood	multiple	30 generators	AP	98.80	mean
[21]	ood	cross-dataset	Chameleon	ACC	62.10	value
[22]	ood	multiple	45 sets	ACC	94.30	mean
[22]	ood	multiple	45 sets	AP	99.20	mean
[22]	robustness	jpeg	JPEG Q30	ACC	86.90	value
[23]	in-domain	none	AIGC-Trace closed set	ACC	98.80	value
[23]	ood	cross-dataset	ForenSynths	ACC	97.30	value
[23]	open-set	open-set	near-OOD rejection	AUC	58.30	value
[24]	in-domain	none	closed set	AUC	100.00	value
[24]	open-set	open-set	fully open (config 1)	AUC	92.00	value
[24]	open-set	open-set	fully open (config 2)	AUC	81.00	value
[24]	open-set	open-set	fully open (config 3)	AUC	85.00	value
[26]	in-domain	none	80/20 split	ACC	91.32	value
[27]	in-domain	none	DFDC	ACC	98.10	value
[30]	ood	unseen-generator	GenImage (SD1.4 protocol)	ACC	77.50	mean
[31]	ood	unseen-generator	GenImage	ACC	80.40	mean
[31]	ood	unseen-generator	DiffusionForensics	ACC	98.70	mean
[32]	in-domain	none	digital validation	ACC	100.00	value
[32]	robustness	print-and-scan	printed-and-scanned	ACC	50.00	value
[33]	open-set	open-set	unknown architectures	AUC	93.58	mean
[34]	in-domain	none	validation split	ACC	98.20	value
[35]	in-domain	none	SIDset	ACC	95.74	value
[35]	ood	unseen-generator	9 external generators	ACC	84.60	mean
[36]	in-domain	none	ADM	ACC	98.00	value
[36]	in-domain	none	ADM	AP	99.90	value
[36]	ood	unseen-generator	earlier generators	ACC	89.90	mean
[36]	ood	unseen-generator	earlier generators	AP	97.70	mean
[36]	ood	unseen-generator	recent generators	ACC	93.60	mean
[36]	ood	unseen-generator	recent generators	AP	99.30	mean
[37]	ood	cross-dataset	Synthbuster	ACC	85.76	value
[37]	ood	multiple	diverse unseen sets	ACC	75.82	value
[38]	ood	cross-dataset	Celeb-DF	AUC	94.25	value
[38]	ood	cross-dataset	DFDC	AUC	78.36	value
[39]	in-domain	none	skin-image dataset	ACC	94.30	value
[40]	in-domain	none	ProGAN	ACC	100.00	value
[40]	ood	unseen-generator	BigGAN	ACC	95.20	value
[40]	ood	unseen-generator	CycleGAN	ACC	98.50	value
[40]	ood	cross-family	GenImage Midjourney	ACC	88.20	value
[40]	ood	cross-family	GenImage SD1.4	ACC	86.50	value
[40]	ood	cross-family	GenImage SD1.5	ACC	92.10	value
[40]	ood	cross-family	GenImage Wukong	ACC	86.30	value
[40]	robustness	jpeg	BigGAN, JPEG Q50	ACC	89.10	value
[40]	robustness	adversarial	BigGAN, adversarial	ACC	83.40	value
[42]	in-domain	none	validation split	ACC	96.25	value
[43]	in-domain	none	internal test	ACC	97.95	value
[46]	in-domain	none	FFHQ/StyleGAN2	ACC	99.20	value
[46]	ood	cross-dataset	CelebA + StyleGAN2	ACC	99.11	mean
[47]	in-domain	none	AIGIBench same-source	ACC	99.70	mean
[47]	ood	unseen-generator	GenImage (SD1.4 protocol)	ACC	96.30	mean
[48]	ood	unseen-generator	UniversalFakeDetect-style benchmark	ACC	95.58	mean
[50]	in-domain	none	ProGAN	AP	100.00	value
[50]	ood	multiple	18 datasets	AP	96.13	mean

Study	Setting	Shift	Target	Metric	Value	Stat
[51]	in-domain	none	FFHQ/StyleGAN	ACC	94.80	value
[51]	ood	unseen-generator	PGAN	ACC	54.75	value
[52]	in-domain	none	seen sources	ACC	95.04	value
[52]	in-domain	none	seen sources	AUC	98.99	value
[52]	ood	cross-dataset	ten unseen subsets	ACC	65.90	mean
[52]	ood	cross-dataset	ten unseen subsets	AUC	77.46	mean
[52]	robustness	jpeg	JPEG Q50	ACC	93.92	value
[55]	ood	multiple	16 generators	ACC	96.13	mean
[55]	ood	multiple	16 generators	AP	99.15	mean
[57]	in-domain	none	DFLIP-3K	ACC	92.30	value
[57]	robustness	jpeg	JPEG Q50	ACC	89.00	value
[58]	in-domain	none	validation	ACC	87.32	value
[58]	ood	unseen-generator	StyleGAN3	ACC	93.00	value
[61]	in-domain	none	DeepGuardDB	ACC	99.87	value
[62]	in-domain	none	ProGAN	ACC	100.00	value
[62]	ood	multiple	16 generators	ACC	94.40	mean
[62]	robustness	jpeg	JPEG	ACC	83.70	value
[63]	ood	multiple	14 categories	ACC	80.82	mean
[63]	robustness	jpeg	JPEG	ACC	79.55	value
[64]	in-domain	none	custom test set	ACC	98.50	value
[65]	in-domain	none	curated collection	ACC	94.80	value
[67]	in-domain	none	SD v1.5 self-test	ACC	71.80	value
[67]	in-domain	none	SD v1.5 self-test	AP	78.50	value
[67]	ood	unseen-generator	leave-one-model-out	ACC	61.20	mean
[67]	ood	unseen-generator	leave-one-model-out	AP	65.10	mean
[68]	in-domain	none	AITfake	ACC	99.56	mean
[68]	ood	cross-dataset	CoCoGLIDE	ACC	54.19	value
[68]	ood	cross-dataset	HIFI-IFDL	ACC	82.97	value
[68]	ood	cross-dataset	GenImage	ACC	80.28	value
[68]	ood	cross-dataset	CASIA	ACC	77.16	value
[69]	in-domain	none	AI-ArtBench	AUC	96.20	value
[70]	in-domain	none	RSI test	ACC	100.00	value
[70]	ood	cross-dataset	ADE20K	ACC	88.90	value
[70]	ood	cross-dataset	Sketchy	ACC	94.30	value
[70]	ood	cross-dataset	CUB-200-2011	ACC	97.80	value
[71]	in-domain	none	ProGAN	ACC	99.95	value
[71]	in-domain	none	ProGAN	AP	100.00	value
[71]	ood	multiple	AIGCDetectBenchmark (16)	ACC	90.97	mean
[71]	ood	multiple	AIGCDetectBenchmark (16)	AP	96.62	mean
[72]	in-domain	none	UFD Benchmark 1	ACC	99.50	value
[72]	in-domain	none	UFD Benchmark 1	AP	99.90	value
[72]	ood	unseen-generator	UFD Benchmark 1 OOD	ACC	93.00	mean
[72]	ood	unseen-generator	UFD Benchmark 1 OOD	AP	97.30	mean
[72]	ood	cross-dataset	Chameleon	ACC	67.83	value
[72]	ood	cross-dataset	Synthbuster	ACC	78.18	mean
[72]	robustness	jpeg	JPEG QF50	ACC	82.79	value
[73]	in-domain	none	in-distribution test	AUC	99.30	value
[73]	ood	multiple	uncompressed OOD	ACC	70.40	mean
[75]	ood	unseen-generator	GAN dataset	ACC	98.80	mean
[75]	ood	cross-family	diffusion dataset	ACC	96.30	mean
[77]	in-domain	none	Celeb-DF	ACC	96.30	value
[78]	in-domain	none	CIFAKE	ACC	96.12	value
[79]	in-domain	none	generic model	MCC	87.20	value
[79]	ood	unseen-generator	generalization training	MCC	52.70	value
[80]	in-domain	none	ProGAN	ACC	99.88	value
[80]	in-domain	none	ProGAN	AP	100.00	value
[80]	ood	multiple	19 unseen sets	ACC	84.57	mean
[80]	ood	multiple	19 unseen sets	AP	92.46	mean
[81]	in-domain	none	20% split	ACC	95.00	value
[82]	in-domain	none	in-domain test	ACC	99.49	value
[82]	ood	cross-dataset	11 cross-dataset domains	ACC	84.88	mean
[82]	robustness	jpeg	JPEG Q30	ACC	81.42	value
[83]	in-domain	none	ADM (LSUN)	ACC	82.70	value
[83]	ood	unseen-generator	4 diffusion generators	ACC	80.40	mean

Study	Setting	Shift	Target	Metric	Value	Stat
[83]	ood	cross-family	StyleGAN	ACC	66.00	value
[84]	in-domain	none	custom split	ACC	98.50	value
[85]	ood	unseen-generator	Tgan	AP	97.40	mean
[86]	in-domain	none	ProGAN	ACC	98.70	value
[86]	in-domain	none	ProGAN	AP	99.90	value
[86]	ood	multiple	38 unseen generators	ACC	87.20	mean
[86]	ood	multiple	38 unseen generators	AP	94.60	mean
[87]	in-domain	none	SD v1.5	AUC	99.02	value
[87]	ood	unseen-generator	10 unseen generators	AUC	90.29	mean
[88]	in-domain	none	CIFAKE	ACC	96.60	value
[90]	in-domain	none	CIFAKE	ACC	98.91	value
[91]	in-domain	none	WILD closed set	AUC	98.13	value
[91]	open-set	open-set	WILD open set	AUC	100.00	value
[92]	in-domain	none	PRISM full generation	ACC	96.60	mean
[92]	ood	cross-dataset	GenImage	ACC	90.30	mean
[92]	ood	cross-dataset	DRCT-2M	ACC	97.40	mean
[94]	ood	cross-dataset	Celeb-DF	AUC	79.20	value
[94]	ood	cross-dataset	DFDC	AUC	73.60	value
[95]	ood	unseen-generator	MMFR (10 generators)	ACC	94.06	mean
[95]	ood	cross-dataset	LOKI	ACC	86.73	value
[97]	in-domain	none	held-out split (binary)	ACC	97.70	value
[97]	ood	multiple	generalization sets	ACC	67.40	mean
[97]	robustness	jpeg	JPEG QF50	ACC	69.85	value
[98]	in-domain	none	ProGAN	ACC	99.50	value
[98]	in-domain	none	ProGAN	AP	100.00	value
[98]	ood	unseen-generator	cross-GAN	ACC	92.90	mean
[98]	ood	unseen-generator	cross-GAN	AP	98.50	mean
[98]	ood	cross-family	cross-diffusion	ACC	89.50	mean
[98]	ood	cross-family	cross-diffusion	AP	97.70	mean

S3 Full-Text Exclusions and Reports Not Retrieved

Table S3 lists the 7 reports excluded at full-text assessment, with the exclusion code defined in the Review Methodology section of the main paper. Table S4 lists the 39 reports that were sought but could not be retrieved through institutional access.

Table S3: Reports excluded at full text

Record	Title	Code
R0028	Training humans for synthetic face image detection	FT-EC5
R0061	Design of an integrated model combining vision transformer, multi-modal fusion,	FT-EC9
R0123	Generative Adversarial Ensemble Learning for Face Forensics	FT-EC4
R0130	ADVANCED DEEPFAKE DETECTION LEVERAGING SWIN TRANSFORMER TECHNOLOGY	FT-EC4
R0151	Hybrid Deep Learning for Image Authenticity: Distinguishing Between Real and AI-	FT-EC3
R0217	Deepfake Video Forensics: A Hybrid Deep Learning Model for Detecting Deepfake Fa	FT-EC14
R0329	An efficient deepfake video detection using robust deep learning	FT-EC14

Table S4: Reports sought but not retrieved

Record	Title	Year
R0007	AI-generated image detection algorithm based on classical-quantum hybrid neural network	2026
R0015	ImageAuthNet: toward an explainable and generalizable framework to detect fake images - use cas	2026
R0022	DADP: a discrepancy-aware dual-path network for AI-generated image detection	2026
R0026	Semantic Distribution and Authenticity Discrepancy Alignment for AI-Generated Image Detection	2026
R0043	Image quality assessment based fake face detection	2023
R0054	Bi-LORA: A Vision-Language Approach for Synthetic Image Detection	2025
R0059	Research on adversarial identification methods for AI-generated image software Craiyon V3	2025
R0060	Self-Supervised AI-Generated Image Detection: A Camera Metadata Perspective	2026
R0075	ArtGate: Injecting Fake Artifact Features into CLIP for AI-Generated Image Detection	2026
R0079	Cross-generator image authenticity verification: multi-modal feature fusion and distribution di	2026
R0084	SPRITZ-PS: validation of synthetic face images using a large dataset of printed documents	2024
R0087	Toward Generalizable AI-Generated Image Detection with a New Realistic Dataset: Performance Eva	2026
R0088	Source Attribution of AI-Generated Images in Data Scarcity Conditions via Training-Free Resynth	2026
R0100	High-resolution network-based multi-feature fusion for generalized forgery detection	2025
R0107	AI-Generated Image Detection Method Integrating Light -Shadow Sensitive Features and Kolmogorov	2025
R0112	MSAFNet: multi-scale self-adaptive feature fusion network for AI-generated image detection	2025
R0116	Synthetic image detection method integrating neural ordinary differential equations and spatial	2026
R0117	Transferable Dual-Domain Feature Importance Attack Against AI-Generated Image Detector	2026
R0119	ERASE: Bypassing Collaborative Detection of AI Counterfeit via Comprehensive Artifacts Eliminat	2026
R0125	CoDA: Color Distribution Probing for Efficient and Generalizable AI-Generated Image Detection	2026
R0127	Hybrid AI approaches for detecting deepfake faces	2025
R0129	Quality-Guided Forgery Adapter for Generalizable AIGC Image Detection	2026
R0132	Multi-Domain Feature Fusion: Leveraging Color, Noise, and Spatial Features for AI-generated Ima	2026
R0134	Adversarial Mixture of Experts Post-Training for Robust AI-Generated Image Detection	2026
R0138	Dual Frequency Branch Framework With Reconstructed Sliding Windows Attention for AI-Generated I	2026
R0139	FALCON-Net: Feature Aggregation of Local Patterns for AI-Generated Image Detection	2026
R0140	Dual-scale model collaborative reasoning with multi-feature fusion for robust AI-generated imag	2026
R0141	Towards Extensible Detection of AI-Generated Images via Content-Agnostic Adapter-Based Category	2025
R0142	Optimized Frequency Collaborative Strategy Drives AI Image Detection	2025
R0143	Enhancing deep fake detection: comparative analysis of advanced deep learning architectures	2026
R0145	FAMSeC: A Few-Shot-Sample-Based General AI-Generated Image Detection Method	2025
R0149	Wavelet-packets for deepfake image analysis and detection	2022
R0150	Learning a Semantic Similarity Orthogonal Space for Model-Level AI-Generated Image Source Attri	2026
R0153	Incremental Learning for AI-Generated Image Detection	2026
R0222	FD-GAN: Generalizable and Robust Forgery Detection via Generative Adversarial Networks	2024
R0252	Beyond Familiar Domains: A Study of the Generalization Ability of Machine-Generated Image Detec	2025
R0265	RealDTT: Towards A Comprehensive Real-World Dataset for Tampered Text Detection	2025
R0301	Exploring Bi-Level Inconsistency via Blended Images for Generalizable Face Forgery Detection	2024
R0339	Enhancing Deepfake Detection: Leveraging StyleGAN3 for Robust AI-Generated Forgery Identificati	2026

S4 Scopus Search Strings

The two searches were run on Scopus on 13 August 2026 (title, abstract and keywords; articles in English, 2020–2026; no subject-area restriction).
S01 (primary):

```
TITLE-ABS-KEY( ( "AI-generated image*" OR "AI generated image*" OR "synthetic image*" OR "machine-generated image*" OR "generative AI image*" ) AND ( "image detection" OR "synth
```

S02 (generalization-focused):

```
TITLE-ABS-KEY( ( "AI-generated image*" OR "AI generated image*" OR "synthetic image*" OR "machine-generated image*" OR "generative AI image*" ) AND ( detect* OR forensic* OR aut
```

References

[1] P. Naga Srinivas and M. Chandra Sekhara Rao, "A Multimodal Visionlanguage Model for Shrimp Species Classification and AI-generated Image Detection Using CLIP and BERT," *International Journal of Intelligent Engineering and Systems*, vol. 19, no. 8, pp. 425–441, 2026.

[2] S. Zhao, W. Liu, D. Xu, J. Zhou, and Z. Wang, "PhyTrace: Tracing Physical Inconsistency in AI-Generated Images via ISP Emulation," *IEEE Transactions on Image Processing*, vol. 35, pp. 8009–8020, 2026.

[3] P. Sharma, M. Kumar, and H. Sharma, "Robust GAN-Based CNN Model as Generative AI Application for Deepfake Detection," *EAI Endorsed Transactions on Internet of Things*, vol. 10, 2024.

[4] V. Bhamidipati, S. Chanamallu, and S. Gopinathan, "A Multi-Branch Transformer Based Cross-Attention Framework for Computer-Generated Image Detection," *International Journal of Image, Graphics and Signal Processing*, vol. 18, no. 4, pp. 25–47, 2026.

[5] G.-H. Lin, Y.-T. Lin, and J.-L. Chen, "Synthetic Image Detection Based on Multi-Level Feature Fusion and Data Augmentation," *ICT Express*, 2026.

[6] K. Liang, P. Lu, X. Liu, X. Lu, and A. Li, "SCA-Det: A structure-context artifact detector for AI-generated image detection," *Neurocomputing*, vol. 685, p. 133627, 2026.

[7] W. Qu, C. Chen, W. Lu, and T. Li, "MS-DCL: Multi-Scale Decoupled Contrastive Learning for universal AI-generated image detection," *Knowledge-Based Systems*, vol. 346, p. 116183, 2026.

[8] M. Alam, M. Golam, and T. Jun, "FMACNN: Federated multi-attention CNN framework for artificial image detection," *Journal of Information Security and Applications*, vol. 100, p. 104462, 2026.

[9] J. Chen, M. Lo, H. Liao, and T. Huang, "IPD-Net: Detecting AI-Generated Images via Inter-Patch Dependencies," *International Journal of Advanced Computer Science and Applications*, vol. 15, no. 7, 2024.

[10] Y. Wang, Z. Huang, Z. Su, A. Prugel-Bennett, and X. Hong, "Penny-Wise and Pound-Foolish in AI-Generated Image Detection," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 48, no. 7, pp. 7707–7719, 2026.

[11] Y. Diao, N. Zhai, C. Miao, Z. Yu, X. Wei, X. Yang, and M. Wang, "Vulnerabilities in AI-Generated Image Detection: The Challenge of Adversarial Attacks," *IEEE Transactions on Multimedia*, vol. 28, pp. 8709–8721, 2026.

[12] T. Ghosh and R. Naskar, "A robust synthetic face detector in OSN context based on Gradient of Color features," *Forensic Science International*, vol. 380, p. 112817, 2026.

[13] K. Sreeraj and B. Arun Kumar, "Transfer Learning-Driven Hybrid Architecture for Deepfake Detection Using InceptionResNetV2 and Stacked Gated Convolutional Units," *International Journal of Intelligent Engineering and Systems*, vol. 18, no. 8, pp. 230–252, 2025.

[14] S. Raj, J. Mathew, and A. Mondal, "Generalized and robust model for GAN-generated image detection," *Pattern Recognition Letters*, vol. 182, pp. 104–110, 2024.

[15] Y. Shi, S. Weng, L. Yu, and R. Ni, "DCNet: Learning Similarity and Spatial Complementary Features for Generalized AI-Generated Image Detection," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 36, no. 6, pp. 8717–8731, 2026.

[16] J. Wang, B. Tondi, and M. Barni, "An Eyes-Based Siamese Neural Network for the Detection of GAN-Generated Face Images," *Frontiers in Signal Processing*, vol. 2, 2022. Art. no. 918725.

[17] Y. Tang, X. Liu, X. Ding, H. Guo, Y. Wang, and H. Deng, "SF-CFNet: Spatial-frequency collaborative fusion detection network for AI-generated satellite images," *Expert Systems with Applications*, vol. 330, p. 133006, 2026.

[18] A. Hamadene and M. Allili, "Cross-Model Deepfake Detection Through Contourlet-Based Inter-Channel Spectral Analysis," *IEEE Transactions on Biometrics, Behavior, and Identity Science*, vol. 8, no. 6, pp. 989–1000, 2026.

[19] H. Zen, R. Wagh, M. Wanderley, G. Bicalho, R. Park, M. Sun, R. Palacios, L. Carvalho, G. Rinaldo, and A. Gupta, "Ensemble-Based Biometric Verification: Defending Against Multi-Strategy Deepfake Image Generation," *Computers*, vol. 14, no. 6, p. 225, 2025.

[20] R. Yumlembam, B. Issac, N. Aslam, E. Babu, J. Collyer, and F. Kennedy, "Detection of AI generated images using combined uncertainty measures and particle swarm optimised rejection mechanism," *Scientific Reports*, vol. 15, no. 1, 2025. Art. no. 44021.

[21] H. Lin, J. Qin, X. Wu, T. Chen, and Z. Yang, "Revisiting DIRE: towards universal AI-generated image detection," *Neural Networks*, vol. 193, p. 108084, 2026.

[22] P. Lu, K. Liang, X. Liu, X. Lu, and A. Li, "SDA-Det: A subpixel-aware dual-domain artifact detector for AI-generated image detection," *Expert Systems with Applications*, vol. 322, p. 132304, 2026.

[23] Z. Qin, S. Gu, Y. Zhang, L. Xie, M. Jing, Q. Xiang, Y. Guo, Z. Wang, and Y. Hu, "BM-DDFN: Bilinear cross-domain modeling for AIGC image source attribution," *Knowledge-Based Systems*, vol. 340, p. 115655, 2026.

[24] L. Abady, J. Wang, B. Tondi, and M. Barni, "A siamese-based verification system for open-set architecture attribution of synthetic images," *Pattern Recognition Letters*, vol. 180, pp. 75–81, 2024.

[25] D. Coccomini, A. Esuli, F. Falchi, C. Gennaro, and G. Amato, "Detecting images generated by diffusers," *PeerJ Computer Science*, vol. 10, p. e2127, 2024.

[26] F. Alrowais, A. Hassan, W. Almukadi, M. Alanazi, R. Marzouk, and A. Mahmud, "Boosting Deep Feature Fusion-Based Detection Model for Fake Faces Generated by Generative Adversarial Networks for Consumer Space Environment," *IEEE Access*, vol. 12, pp. 147680–147693, 2024.

[27] Q. Man and Y.-I. Cho, "Transformer Based on Multi-Domain Feature Fusion for AI-Generated Image Detection," *Electronics (Switzerland)*, vol. 15, no. 3, p. 716, 2026.

[28] S. Madan, Y. Yasuda, and J. Yamagishi, "VeriTrace: A Multitask Disaster Dataset for RealFake Verification and Source Tracing," *IEEE Open Journal of Signal Processing*, vol. 7, pp. 438–446, 2026.

[29] S. Klier and H. Baier, "Boon or Bane: Source Camera Identification meets AI-generated images," *Forensic Science International: Digital Investigation*, vol. 56, p. 302064, 2026.

[30] V.-N. Tran, P. Choi, H.-S. Le, S.-H. Lee, and K.-R. Kwon, "DiffCoR: Exposing AI-Generated Image by Using Stable Diffusion Model Based on Consistent Representation Learning," *IEEE Open Journal of the Computer Society*, vol. 6, pp. 1353–1365, 2025.

[31] Z. Meng, B. Peng, J. Dong, T. Tan, and H. Cheng, "Artifact feature purification for cross-domain detection of AI-generated images," *Computer Vision and Image Understanding*, vol. 247, p. 104078, 2024.

[32] A. Ferreira, E. Nowroozi, and M. Barni, "VIPPrint: Validating Synthetic Image Detection and Source Linking Methods on a Large Scale Dataset of Printed Documents," *Journal of Imaging*, vol. 7, no. 3, p. 50, 2021.

[33] J. Wang, B. Tondi, and M. Barni, "BOSC: A Backdoor-Based Framework for Open Set Synthetic Image Attribution," *IEEE Transactions on Information Forensics and Security*, vol. 20, pp. 8043–8058, 2025.

[34] D. Lamichhane, "Advanced Detection of AI-Generated Images Through Vision Transformers," *IEEE Access*, vol. 13, pp. 3644–3652, 2025.

[35] L. Li, T. Lu, J. Song, and K. Cheng, "SIDWA: Synthetic Image Detection Based on Discrete Wavelet Transform Stem and Deformable Sliding Window Cross-Attention," *Electronics (Switzerland)*, vol. 15, no. 4, p. 891, 2026.

[36] J. Li, W. Ou, D. Meng, and Y. Xu, "Background-Noise-Driven Detection of Diffusion-Generated Images," *IEEE Signal Processing Letters*, vol. 33, pp. 1671–1675, 2026.

[37] J. Deng, C. Zhang, C. Xu, C. Lin, Z. Zhao, S. Liu, Q. Wang, and C. Shen, "CLIP-ADA: CLIP-Guided Artifact-Invariant Generalizable Synthetic Image Detection," *IEEE Transactions on Information Forensics and Security*, vol. 21, pp. 3588–3601, 2026.

[38] Q. Liu, Z. Xue, H. Liu, and J. Liu, "Enhancing Deepfake Detection With Diversified Self-Blending Images and Residuals," *IEEE Access*, vol. 12, pp. 46109–46117, 2024.

[39] A. Shahzad, B. Zafar, R. Talib, and M. Khan, "A hybrid transfer learning approach for multi-class classification of medical deepfakes generated by GAN and diffusion models," *Discover Computing*, vol. 29, no. 1, 2026. Art. no. 225.

[40] H. Yu and B. Xu, "Multi-modal texture fusion network for detecting AI-generated images," *Frontiers in Artificial Intelligence*, vol. 8, 2025. Art. no. 1663292.

[41] J. Chory and D. ., "AI-generated image detection using Machine Learning techniques," *International Journal of Electronics and Telecommunications*, vol. 72, no. 2, pp. 1–7, 2026.

[42] H. Wang, "Vision Transformer-Based Framework for AI-Generated Image Detection in Interior Design," *Informatica (Slovenia)*, vol. 49, no. 16, 2025.

[43] B. Deepa, C. Lokesh, D. Umamaheswari, B. Ayshwarya, P. Yethish, and K. Suhaas, "An enhanced deep learning framework for DeepFake detection using EfficientNet-B3 comparative evaluation of deep and machine learning techniques," *Discover Computing*, vol. 29, no. 1, 2026. Art. no. 18.

[44] M.-L. Wu and C.-P. Cheng, "Sanitizing Diffusion-Generated Images via Fingerprint Removal and Adversarial Perturbation for Forensic Evasion," *IEEE Access*, vol. 13, pp. 156843–156854, 2025.

[45] S. Iliopoulou, P. Tsinganos, D. Ampeliotis, and A. Skodras, "Synthetic Face Discrimination via Learned Image Compression," *Algorithms*, vol. 17, no. 9, p. 375, 2024.

[46] T. Ghosh and R. Naskar, "Less is more: A minimalist approach to robust GAN-generated face detection," *Pattern Recognition Letters*, vol. 179, pp. 185–191, 2024.

[47] J. Li, M. Zhang, S. Huang, X. Li, P. Feng, and L. Liu, "Geometric-semantic dual-adaptation for generalizable AI-generated image detection," *Neurocomputing*, vol. 700, p. 134418, 2026.

[48] J. Li and K. Wang, "Building a Universal Detector of AI-Generated Images Without Training on Them," *International Journal of Intelligent Systems*, vol. 2025, no. 1, 2025. Art. no. 8530953.

[49] Y. Wang, X. Hong, and Z. Huang, "Benchmark dataset and framework for continual AI-generated image detection," *Journal of Image and Graphics*, vol. 30, no. 11, pp. 3438–3450, 2025.

[50] Y. Lin, T. Mao, Z. Chen, H. Lu, Z. Chen, and Y. Kang, "MFF-Net: A multi-view feature fusion network for generalized forgery image detection," *Neurocomputing*, vol. 640, p. 130351, 2025.

[51] S. Kanwal, S. Tehsin, and S. Saif, "EXPOSING AI GENERATED DEEPFAKE IMAGES USING SIAMESE NETWORK WITH TRIPLET LOSS," *Computing and Informatics*, vol. 41, no. 6, pp. 1541–1562, 2022.

[52] Z. Liu, X. Wang, Q. Wang, N. Bai, R. Han, and S. Pang, "Structure searchable network model for deepfake image detection," *Pattern Recognition*, vol. 179, p. 113813, 2026.

[53] S. Ali Raza, U. Habib, M. Usman, A. Ashraf Cheema, and M. Sajid Khan, "MMGANGuard: A Robust Approach for Detecting Fake Images Generated by GANs Using Multi-Model Techniques," *IEEE Access*, vol. 12, pp. 104153–104164, 2024.

[54] A. Panda, T. Ghosh, T. Choudhary, and R. Naskar, "Bias-Free? An Empirical Study on Ethnicity, Gender, and Age Fairness in Deepfake Detection," *ACM Computing Surveys*, vol. 58, no. 10, pp. 1–29, 2026.

[55] S. Weng, R. Li, L. Yu, G. Yang, and P.-W. Tsai, "DTBF: Combining Local Statistical Artifacts and Concept Alignment for Synthetic Image Detection," *IEEE Signal Processing Letters*, vol. 33, pp. 2834–2838, 2026.

[56] D. Park, H. Na, and D. Choi, "Performance Comparison and Visualization of AI-Generated-Image Detection Methods," *IEEE Access*, vol. 12, pp. 62609–62627, 2024.

[57] Y. Wang, X. Hong, Y. Li, Z. Ma, and Z. Huang, "Linguistic profiling of deepfakes: An open database for next-Generation deepfake detection," *Pattern Recognition*, vol. 178, p. 113395, 2026.

[58] G. Rozhbayani, A. Tuama, and H. Abdulkhudhur, "Fusing Frequency and Spatial Transformers for Robust Detection of AI-Generated Images," *Ingenierie des Systemes d'Information*, vol. 30, no. 10, pp. 2719–2725, 2025.

[59] J. Lu, Z. Fu, N. Zhao, L. Xing, X. Wang, K. Chen, W. Zhang, and N. Yu, "LiteUpdate: A Lightweight Framework for Updating AI-Generated Image Detectors," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 36, no. 8, pp. 11788–11801, 2026.

[60] H. Wang, J. Lan, Y. Kang, H. Zhu, W. Wang, Z. Zhang, and S. Wang, "Generalizable and Adaptive Continual Learning Framework for AI-Generated Image Detection," *IEEE Transactions on Multimedia*, vol. 28, pp. 2693–2704, 2026.

[61] Y. Namani, I. Reghioua, G. Bendiab, M. Labiod, and S. Shiaeles, "DeepGuard: Identification and Attribution of AI-Generated Synthetic Images," *Electronics (Switzerland)*, vol. 14, no. 4, p. 665, 2025.

[62] J. Xu, Y. Du, L. Lyu, and C. Yang, "Harnessing attention for cropping and fusion in CLIP-based AIGC detection," *Neurocomputing*, vol. 669, p. 132482, 2026.

[63] H. Yang, R. Ma, G. Wang, and J. Kato, "NSFF: Noise and semantic features fusion for AI-generated image detection," *Neural Networks*, vol. 203, p. 109130, 2026.

[64] O. Tariq, M. Arshed, M. Kabir, K. Ijaz, G. .C., and H. Batool, "A Transfer-Learning Approach for Detection of Multiclass Synthetic Skin Cancer Images Generated by Deep Generative Models to Prevent Medical Insurance Fraud," *Mathematical and Computational Applications*, vol. 31, no. 1, p. 31, 2026.

[65] R. Raikwar, A. Ansari, A. Shrivas, A. Ubale, A. Kumar, and S. Kaur, "PHOTO AUTHENTICITY DETECTION USING MACHINE LEARNING FOR DEEPFAKE AND AI-GENERATED CONTENT VERIFICATION," *ShodhKosh: Journal of Visual and Performing Arts*, vol. 6, no. 4s, 2025.

[66] Z. Lou, G. Cao, and M. Lin, "Black-box attack against GAN-generated image detector with contrastive perturbation," *Engineering Applications of Artificial Intelligence*, vol. 124, p. 106594, 2023.

[67] Y. Zhou, X. Hu, J. Tong, C. Lin, M. Sun, and S. Chen, "ADRD: Detecting diffusion-generated images via adversarial perturbation induced reconstruction discrepancy," *PloS one*, vol. 21, no. 7, p. e0350655, 2026.

[68] Y. Liu, X. Li, J. Zhang, S. Li, S. Hu, and J. Lei, "Hierarchical Progressive Image Forgery Detection and Localization Method Based on UNet," *Big Data and Cognitive Computing*, vol. 8, no. 9, p. 119, 2024.

[69] X. Zhang, Z. Yu, and J. Zhao, "SCADET: A detection framework for AI-generated artwork integrating dynamic frequency attention and contrastive spectral analysis," *PLOS ONE*, vol. 20, no. 11, p. e0336328, 2025.

[70] S. Baraheem and T. Nguyen, "AI vs. AI: Can AI Detect AI-Generated Images?," *Journal of Imaging*, vol. 9, no. 10, p. 199, 2023.

[71] S. Li, L. Li, Y. Ren, X. Zhang, and G. Feng, "Lightweight AI-Generated image detection based on enhanced common artifact features," *Expert Systems with Applications*, vol. 301, p. 130500, 2026.

[72] J. Lin, Q. Zhou, and J. Li, "Dual-path AI-generated image detection: Leveraging texture-rich and texture-poor patches with global semantic features," *Digital Signal Processing: A Review Journal*, vol. 171, p. 105840, 2026.

[73] A. Maier and C. Riess, "Reliable Out-of-Distribution Recognition of Synthetic Images," *Journal of Imaging*, vol. 10, no. 5, p. 110, 2024.

[74] T. Qiao, H. Shao, S. Xie, and R. Shi, "Unsupervised Generative Fake Image Detector," *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 34, no. 9, pp. 8442–8455, 2024.

[75] X. Li, M. Xiao, H. Wu, Q. Chang, X. Li, and Y. Zhao, "Beyond Causal Models: Multi-Scale Linear Local Attention for AI-Generated Image Detection," *IEEE Signal Processing Letters*, vol. 33, pp. 3068–3072, 2026.

[76] S. Falcón-López, L. Tobarra, A. Robles-Gómez, and R. Pastor-Vargas, "Forensic Analysis of Manipulated Images and Videos," *Applied Sciences (Switzerland)*, vol. 15, no. 23, p. 12664, 2025.

[77] Q. Man, S.-J. Gee, and Y.-I. Cho, "Multi-Domain Perception Transformer for Generalized Forgery Image Detection," *Applied Sciences (Switzerland)*, vol. 16, no. 1, p. 533, 2026.

[78] A. Gunukula, H. Das Gupta, and V. Sheng, "Detecting AI-Generated Images Using a Hybrid ResNet-SE Attention Model," *Applied Sciences (Switzerland)*, vol. 15, no. 13, p. 7421, 2025.

[79] Q. Bammey, "Synthbuster: Towards Detection of Diffusion Model Generated Images," *IEEE Open Journal of Signal Processing*, vol. 5, pp. 1–9, 2024.

[80] H. Xu, Z. Chen, H. Zhang, L. Xue, and H. Zhang, "GCS-Net: A universal AI-generated visual content detection method based on CLIP," *Knowledge-Based Systems*, vol. 323, p. 113806, 2025.

[81] D. Javiya, G. Jethava, H. Parmar, and S. Jethava, "Enhancing deepfake detection with Adaptive-DCGAN and Lite-CNN: a novel approach to image classification," *Discover Applied Sciences*, vol. 7, no. 11, 2025. Art. no. 1229.

[82] H. Shi, M. Liu, and C. Gong, "EXACT: An Explainable Adaptive SemantiC-Trace framework for AI-generated image forensics," *Computer Vision and Image Understanding*, vol. 271, p. 104871, 2026.

[83] Y. Huang, Z. Gu, P. He, C. Li, J. Gou, and X. Lu, "Diffusion-Generated Image Detection via Targeted Mask-Based Knowledge Distillation," *IET Image Processing*, vol. 20, no. 1, 2026. Art. no. e70425.

[84] S. Ishida, Y. Matsuda, J. Kuroiwa, and T. Odaka, "Integrating semantic structure and texture cues for detecting landscape images generated by stable diffusion-based models," *Nonlinear Theory and its Applications, IEICE*, vol. 17, no. 3, pp. 1015–1028, 2026.

[85] H. Wu, J. Zhou, and S. Zhang, "Generalizable Synthetic Image Detection via Language-Guided Contrastive Learning," *IEEE Transactions on Artificial Intelligence*, vol. 7, no. 6, pp. 3485–3496, 2026.

[86] T. Ghosh and R. Naskar, "Spatial flatness-curvature mask driven generalized detection of synthetic images," *Pattern Recognition Letters*, vol. 205, pp. 100–108, 2026.

[87] S. Lu, J. Tian, Y. Zhang, F. Wu, L. Gong, and H. Pan, "DIFC-Net: Diffusion-Intrinsic Feature Capture for AI-Generated Image Detection," *Sensors*, vol. 26, no. 8, p. 2389, 2026.

[88] O. Çnar and Y. Doan, "Novel Deepfake Image Detection with PV-ISM: Patch-Based Vision Transformer for Identifying Synthetic Media," *Applied Sciences (Switzerland)*, vol. 15, no. 12, p. 6429, 2025.

[89] H. Wang, X. Liu, Z. Lu, C. Yan, X. Jiang, R. Wang, and E. Gavves, "RADAR: Reasoning AI-Generated Image Detection for Semantic Fakes," *Technologies*, vol. 13, no. 7, p. 280, 2025.

[90] F. Bayram, "Color Space Analysis for AI-Generated Image Detection on CIFAKE: CNN Models and Statistical Evaluation," *IEEE Access*, vol. 14, pp. 30747–30757, 2026.

[91] C. Giusti, L. Guarnera, and S. Battiato, "Proto-LeakNet: Towards signal-leak aware attribution in synthetic human face imagery," *Computer Vision and Image Understanding*, vol. 270, p. 104848, 2026.

[92] F. Bartolucci, S. Salti, and G. Lisanti, "The PRISM benchmark: PhotoRealistic Image Synthesis and Manipulation to detect generated images," *Computer Vision and Image Understanding*, vol. 270, p. 104826, 2026.

[93] A. Mallet, R. Cogranne, M. Kuribayashi, and A. Méreur, "How Much is the Source Mismatch an Important Problem for Deepfake Detection?," *APSIPA Transactions on Signal and Information Processing*, vol. 14, no. 3, pp. 1–33, 2025.

[94] W. Li and Z. Shen, "FD2Foremer: Thinking Face Forgery Detection in Midfrequency Geometry Details," *Security and Communication Networks*, vol. 2022, pp. 1–15, 2022.

[95] Y. Gao, D. Chang, B. Yu, H. Qin, M. Diao, L. Chen, K. Liang, and Z. Ma, "Toward Generalizable Forgery Detection and Reasoning," *IEEE Transactions on Image Processing*, vol. 35, pp. 3395–3410, 2026.

[96] S. Sharma, A. AlEnizi, M. Kumar, O. Alfarraj, and M. Alowaidi, "Detection of real-time deep fakes and face forgery in video conferencing employing generative adversarial networks," *Heliyon*, vol. 10, no. 17, p. e37163, 2024.

[97] O. Pontorno, L. Guarnera, and S. Battiato, "DeepFeatureX-SN: Generalization of deepfake detection via contrastive learning," *Multimedia Tools and Applications*, vol. 84, no. 39, pp. 47721–47740, 2025.

[98] X. Chen, T. Huang, W. Liu, Z. Wang, W. Li, W. Huang, R. Chen, and H. Luo, "DeepfakeCLIP: Semantic-Opposite prompt learning for generalizable deepfake detection," *Knowledge-Based Systems*, vol. 330, p. 114681, 2025.

[99] A. Agarwal, M. Vatsa, R. Singh, and N. Ratha, "Parameter agnostic stacked wavelet transformer for detecting singularities," *Information Fusion*, vol. 95, pp. 415–425, 2023.